

1.9 More exercises

Example 14. Define an appropriate language and formalize the following sentences using First Order Logic formulas.

1. *Bill has at least one sister.*
2. *Bill has no sister.*
3. *Bill has at most one sister.*
4. *Bill has (exactly) one sister.*
5. *Bill has at least two sisters.*
6. *Every student takes at least one course.*
7. *Only one student failed Geometry.*
8. *No student failed Geometry but at least one student failed Analysis.*
9. *Every student who takes Analysis also takes Geometry.*

Example 15. Consider the following sentences:

1. *All actors and journalists invited to the party are late.*
2. *There is at least a person who is on time.*
3. *There is at least an invited person who is neither a journalist nor an actor.*

Formalize the sentences and prove that 3. is not a logical consequence of 1. and 2.

Example 16. Let $\{c_1, \dots, c_k\}$ be a non empty and finite set of colors. A partially colored directed graph is a structure $\langle N, R, C \rangle$ where

- N is a non empty set of nodes
- R is a binary relation on N
- C associates colors to nodes (not all the nodes are necessarily colored, and each node has at most one color)

Provide a first order language and a set of axioms that formalize partially colored graphs. Show that every model of this theory correspond to a partially colored graph, and vice-versa. For each of the following properties, write a formula which is true in all and only the graphs that satisfies the property:

1. *connected nodes don't have the same color*
2. *the graph contains only 2 yellow nodes*
3. *starting from a red node one can reach in at most 4 steps a green node*
4. *for each color there is at least a node with this color*
5. *the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color*

Problem 3. Minesweeper is a single-player computer game invented by Robert Donner in 1989. The object of the game is to clear a minefield without detonating a mine. The game screen consists of a rectangular field of squares. Each square can be cleared, or uncovered, by clicking on it. If a square that contains a mine is clicked, the game is over. If the square does not contain a mine, one of two things can happen: (1) A number between 1 and 8 appears indicating the amount of adjacent (including diagonally-adjacent) squares containing mines, or (2) no number appears; in which case there are no mines in the adjacent cells. An example of game situation is provided in the following figure:

Provide a first order language that allows to formalize the knowledge of a player in a game state. In such a language you should be able to formalize the following knowledge:

1. there are exactly n mines in the minefield
2. if a cell contains the number 1, then there is exactly one mine in the adjacent cells.



3. show by means of deduction that there must be a mine in the position (3,3) of the game state of picture 3.1

Suggestion: define the predicate $Adj(x, y)$ to formalize the fact that two cells x and y are adjacent

Problem 4. *Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:*

1. *There is no direct connection from Rome to Trento*
2. *There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.*
3. *Regional trains connect towns in the same region*
4. *Intercity trains don't stop in small towns.*

Bibliography

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